

No. of Functional Features Included in the Solution

Date	26 June 2026
Team ID	SWTID-2026-6095
Project Name	EduGenie
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Educational Concept Explainer	Takes complex topics from users and generates simplified, student-friendly text using a locally hosted LLM pipeline (LaMini-Flan-T5-248M).	Explanation_module.py	Done	
2	Educational Text Summarization	Processes large educational paragraphs and extracts concise, bulleted revision summaries using cloud-based Google Gemini 1.5 pro API with asynchronous handling.	summary_module.py	Done	
3	Quiz Generator	Generates contextual evaluation questions and quiz options based on the provided learning material.	quiz_module.py	Done	
4	Learning Path	This feature enables the application to dynamically architect a step-by-step, sequential educational roadmap based on a target topic entered by the student.	learning_path.py	Done	
5	Intelligent qna	Generates contextual answers.	qna.py	Done	
6	Single Page UI	A clean, lightweight frontend dashboard designed entirely using semantic HTML and custom CSS	Static/index.html	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	6

Total Features Implemented	6
Core / Must-Have Features	5
Additional / Nice-to-Have Features	1
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	Responsive single-page web app using Jinja2 templates, structured CSS styling, dynamic loaders, and interactive quiz/Q&A layouts.
2	Backend / Logic	1	FastAPI web application routing user queries, parsing models, validating parameters via Pydantic, and rendering HTML templates.
3	Database / Storage	0	Stateless system configuration; does not use databases. Secrets are loaded via python-dotenv from project root environment file.
4	API / Integration	5	Google Generative AI SDK (Gemini 1.5 pro) integration for Q&A, Quizzes, Summaries, Roadmaps; Transformers pipeline (LaMini) locally.
5	Security / Authentication	1	Secure storage of Google API credentials in env variables on server-side, preventing leakage to client-side static assets.